

Akash Bhat

US Citizen | Madison, WI | +1 (608) 640-1689
aabhat@wisc.edu | [linkedin.com/in/akashbhat1](https://www.linkedin.com/in/akashbhat1) | github.com/quentesia

Education

University of Wisconsin Madison – BS in Computer Science, Data Science Expected May 2026
Dean's List — Fall 2025, Fall 2024, Spring 2024, Fall 2023 **3.700 GPA**
Relevant Coursework: Algorithms, OS, DBMS, Parallel & Throughput Optimization

Experience

Software Engineering Intern, Skylark Drones May 2025 – Nov 2025

- Developed algorithm for **Mining Bench Detection** deployed as a **low-latency Python microservice** with **Docker** on **AWS**, reducing processing time by **90%**
- Owned end-to-end service lifecycle including **deployment**, **monitoring**, and **troubleshooting**, achieving **96% accuracy** in production with fault-tolerant architecture
- Engineered **microservice infrastructure** on **AWS/Docker** and developed **CLI**/automation tools with **Python** and **Bash** to streamline build, testing, and release across **Linux/Unix**

Undergraduate Peer Mentor – Programming III, UW Madison Sep 2024 – Present

- Provided weekly tutoring in Java data structures, Object-Oriented Programming, Git version control, and JUnit
- Created supplementary materials and coding challenges incorporating code linting and static analysis

Software Engineering Intern, Numberz.ai July 2023 – August 2023

- Developed **low-latency RESTful web services** using **Python** (FastAPI/Flask) and **AWS Lambda**, achieving **30%** faster API response times through **caching** and query optimization
- Built **large-scale data processing** pipelines with **Python** and **Bash**, processing **500K+** documents via **Apache Spark** with CI/CD-integrated deployment

Skills

Languages: Python, SQL, Java, C++ , Bash, JavaScript, Rust

Backend & Databases: REST APIs, FastAPI, Flask, Django, RabbitMQ, Apache Spark, PostgreSQL, Redis, MongoDB

Cloud & DevOps: AWS (Lambda, EC2, S3), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD Pipelines

Frontend: React, HTML/CSS, JavaScript

Monitoring & Tools: Prometheus, Grafana, Splunk, Git/GitHub, Jira, Agile/Scrum, Unix/Linux

Projects

Anvil – AI-Powered Dependency Manager pypi.org/product/anvil-py

- Published PyPI Package:** anvil-py – Python CLI tool leveraging **LLM APIs** to automate dependency upgrades with intelligent changelog analysis and **AI-driven** breaking change prediction
- Designed extensible **object-oriented architecture** supporting pip, poetry, uv, and conda with safe rollback mechanisms and automated release workflows
- Built comprehensive **CI/CD pipeline** with **Jenkins** and **GitHub Actions** for automated testing, linting (PyLint), and package publishing

Asynchronous ML Model Training Web Service github.com/quentesia/Distributed-ML-Training-Platform

- Tech Stack:** Python, FastAPI, RabbitMQ, Celery, Redis, Docker, Jenkins
- Designed **high-throughput distributed service** with **FastAPI** REST APIs, **RabbitMQ** message queuing, and **Celery** workers, processing **10K batches/day** with **50%** reduced latency via **Redis** caching
- Architected **low-latency microservices** with **CI/CD** (Jenkins) supporting fault-tolerant parallel processing of **5K concurrent tasks** on **Docker/AWS**

Yoinker – Linux Clipboard Daemon github.com/quentesia/yoinker

- Tech Stack:** Rust, Linux, X11/Wayland, systemd, IPC
- Built a **Linux** clipboard persistence daemon in **Rust** with **X11/Wayland** protocol support, **systemd** service integration, and **IPC**-based inter-process communication